Problem Set 1 Solutions

1. [10 Points.] Let's apply Bayes' Rule to a COVID test example. Let X be a random variable indicating whether or not a person has COVID, where $X = 1$ means they do have COVID, and $X = 0$ means they don't. The person has a low prior probability $p(X = 1) = 0.001$ of having COVID. Likewise $Y = 1$ indicates a positive COVID test result, and $Y = 0$ means a negative test result, but the test is not perfect. Suppose the test has a false negative rate $p(Y = 0 \mid X = 1) = 0.01$, and a false positive rate $p(Y = 1 \mid X = 0) = 0.1$.

   (a) [5 Points.] Suppose the person's first test is positive, $Y = 1$. Use Bayes' rule to find the posterior probability of the person having COVID, $p(X = 1 \mid Y = 1)$.

   Solution:

   Using Bayes' Rule:

   $$p(X = 1 \mid Y = 1) = \frac{p(Y = 1 \mid X = 1)p(X = 1)}{p(Y = 1)} \tag{1}$$

   In this problem:

   $$\begin{aligned} p(Y = 1 \mid X = 1) &= 0.99 \\ p(X = 1) &= 0.001 \\ p(Y = 1) &= \sum_X p(Y = 1, X) \\ &= p(Y = 1 \mid X = 0)p(X = 0) + p(Y = 1 \mid X = 1)p(X = 1) \\ &= 0.1 \times 0.999 + 0.99 \times 0.001 \\ &= 0.10089 \end{aligned}$$

   Plug these numbers into (1), yielding

   $$p(X = 1 \mid Y = 1) = \frac{0.99 \times 0.001}{0.10089} = 0.00981.$$

   (b) [5 Points.] Suppose the test is negative, $Y = 0$. Use Bayes' rule to find the posterior probability of having COVID, $p(X = 1 \mid Y = 0)$.

   Solution:

   Similar to part (a), use Baye's Rule and we get:

   $$p(X = 1 \mid Y = 0) = \frac{p(Y = 0 \mid X = 1)p(X = 1)}{p(Y = 0)} \tag{2}$$

   In this problem:

   $$\begin{aligned} p(Y = 0 \mid X = 1) &= 1 - p(Y = 1 \mid X = 1) = 0.01 \\ p(X = 1) &= 0.001 \\ p(Y = 0) &= \sum_X p(Y = 0, X) \\ &= p(Y = 0 \mid X = 0)p(X = 0) + p(Y = 0 \mid X = 1)p(X = 1) \\ &= 0.9 \times 0.999 + 0.01 \times 0.001 \\ &= 0.8991 \end{aligned}$$

   Plug these numbers into (2),

   $$p(X = 1 \mid Y = 0) = \frac{0.01 \times 0.001}{0.8991} = 1.11 \times 10^{-5}.$$

2. [10 Points.] Recall the definition of conditional independence: RVs Y_1 and Y_2 are conditionally independent given a third RV X iff $p(y_1, y_2 \mid x) = p(y_1 \mid x)p(y_2 \mid x)$.

   (a) [5 Points.] Prove that the above definition is equivalent to $p(y_1 \mid y_2, x) = p(y_1 \mid x)$ which is equivalent to $p(y_2 \mid y_1, x) = p(y_2 \mid x)$.

   Solution:

   Start with the given equation:

   $$p(y_1, y_2 \mid x) = p(y_1 \mid x)p(y_2 \mid x)$$

   Use Bayes' Rule conditioned on x,

   $$\begin{aligned} p(y_1 \mid y_2, x) &= \frac{p(y_1, y_2 \mid x)}{p(y_2 \mid x)}, \\ &= \frac{p(y_1 \mid x)p(y_2 \mid x)}{p(y_2 \mid x)}, \\ &= p(y_1 \mid x). \end{aligned}$$

   Since the equations above are all equivalent, we have $p(y_1 \mid y_2, x) = p(y_1 \mid x)$ if and only if $p(y_1, y_2 \mid x) = p(y_1 \mid x)p(y_2 \mid x)$.

   (b) [5 Points.] Consider three independent random variables X, W_1, and W_2. Construct two new random variables $Y_1 = X + W_1$ and $Y_2 = X + W_2$. Are Y_1 and Y_2 independent? Are they conditionally independent given X? Prove your results.

   Solution:

   One possible solution: $Y_1 = X + W_1$, $Y_2 = X + W_2$

   We can easily show Y_1, Y_2 are not independent:

   $$\begin{aligned} (Y_1, Y_2) &= (W_1 + X, W_2 + X), \\ &= (W_1, W_2) + (W_1, X) + (W_2, X) + (X, X), \\ &= (X). \end{aligned}$$

   In general, (Y_1, Y_2) need not equal 0, so Y_1 and Y_2 are not independent[1].

   Now we show that Y_1 and Y_2 are conditionally independent given X:

   $$\begin{aligned} P(Y_1 \mid Y_2, X) &= P(W_1 + X \mid W_2 + X, X), \\ &= P(W_1 + X \mid W_2, X), \\ &= P(W_1 + X \mid X), && \text{as } W_2 \text{ and } W_1 + X \text{ are independent}, \\ &= P(Y_1 \mid X), \end{aligned}$$

   as needed.

[1] The converse of this statement is actually false, i.e., we can have $(Y_1, Y_2) = 0$ without Y_1 and Y_2 being independent.

3. [15 Points.] Consider again the person that may have COVID. Suppose the first test is positive for the illness, but the person wants a second opinion. Call the first test outcome $Y_1 = 1$, and the second test outcome Y_2. Assume X doesn't change between the two tests. Assume the measurements are conditionally independent, conditioned on the state X.

   (a) [5 Points.] Suppose the second test is also positive, $Y_2 = 1$. Use a recursive Bayesian estimator to find the posterior $p(X = 1 \mid Y_1 = 1, Y_2 = 1)$ given your result in Problem 1 above.

   Solution:

   Using our expression for the posterior from Problem 1, applying Bayes' rule again yields

$$\begin{aligned} p(X = 1 \mid Y_1 = 1, Y_2 = 1) &= \frac{p(X = 1 \mid Y_1 = 1)p(Y_2 = 1 \mid X = 1)}{\sum_X p(Y_2 = 1 \mid X)p(X \mid Y_1 = 1)} \\ &= \frac{0.00981 \times 0.99}{0.1 \times 0.99019 + 0.99 \times 0.00981} \\ &= 0.0893 \end{aligned}$$

   (b) [10 Points.] How many consecutive positive tests will it take for the posterior probability of having COVID to be greater than not having COVID, i.e. what is n such that $p(X = 1 \mid Y_{1:n} = (1, \ldots, 1)) > 0.5$?

   Solution:

   We begin by writing an expression for the posterior belief of n positive tests,

$$p(X = 1 \mid Y_{1:n} = (1, \ldots, 1)) = \frac{p(Y_n = 1 \mid X = 1)p(X = 1 \mid Y_{1:n-1} = 1)}{\sum p(Y_n = 1 \mid X)p(X \mid Y_{1:n-1} = 1)}$$

   For simple notation, use TP (true positive) to represent $p(Y = 1 \mid X = 1) = 0.99$, FP (false positive) to represent $p(Y = 1 \mid X = 0) = 0.1$, and p_n to represent $p(X = 1 \mid Y_{1:n} = 1)$.

   Thus,

$$p_n = \frac{TP \times p_{n-1}}{TP \times p_{n-1} + FP(1 - p_{n-1})}$$

   Rearrange,

$$(\frac{1}{p_n} - 1) = \frac{10}{99}(\frac{1}{p_{n-1}} - 1) = (\frac{10}{99})^{n-1}(\frac{1}{p_1} - 1) = 100.937(\frac{10}{99})^{n-1}$$

   Solve for $p_n > 0.5$, $n = 4$. So it will take 4 positive tests.

4. [15 Points.] As time passes, the person may become infected with COVID, or recover from it. Let $(X_1, \ldots, X_T)$ be the health state of the person on T consecutive days, and suppose X_t evolves as a Markov chain, with infection rate $p(X_{t+1} = 1 \mid X_t = 0) = 0.001$ and recovery rate $p(X_{t+1} = 0 \mid X_t = 1) = 0.05$. The student has one test per day. We will formulate a Bayesian filter to determine the probability of having the illness given a history of test results.

   (a) [5 Points.] To compress notation, let $\pi_t := [p(X_t = 0 \mid Y_{1:t} = y_{1:t}) \quad p(X_t = 1 \mid Y_{1:t} = y_{1:t})]^T$ be the belief vector at day t, let

$$T := \begin{bmatrix} p(X_t = 0 \mid X_{t-1} = 0) & p(X_t = 0 \mid X_{t-1} = 1) \\ p(X_t = 1 \mid X_{t-1} = 0) & p(X_t = 1 \mid X_{t-1} = 1) \end{bmatrix}$$

   be the transition matrix, and

$$M(y_t) := \begin{bmatrix} p(Y_t = y_t \mid X_t = 0) & 0 \\ 0 & p(Y_t = y_t \mid X_t = 1) \end{bmatrix}$$

   be the measurement likelihood matrix. Write down the recursive Bayesian filter in matrix form using only $M(y_t)$, T, π_{t-1} and π_t. (Hint: Feel free to use a vector of all ones, $\underline{1}$, if it will simplify your notation.)

   Solution:

   The first term in π_{t+1}:

$$\begin{aligned} p(X_{t+1} = 0 \mid Y_{1:t+1} = y_{1:t+1}) &= \frac{p(Y_{t+1} = y_{y+1} \mid X_{t+1} = 0) p(X_{t+1} = 0 \mid Y_{1:t} = y_{1:t})}{\sum_{X_{t+1}} p(Y_{t+1} = y_{y+1} \mid X_{t+1} = 0) p(X_{t+1} = 0 \mid Y_{1:t} = y_{1:t})} \\ &= \frac{p(Y_{t+1} = y_{y+1} \mid X_{t+1} = 0) T(1,:) \pi_t}{\sum_{X_{t+1}} p(Y_{t+1} = y_{y+1} \mid X_{t+1} = 0) p(X_{t+1} = 0 \mid Y_{1:t} = y_{1:t})} \end{aligned}$$

   Similarly, the second term in π_{t+1}:

$$\begin{aligned} p(X_{t+1} = 1 \mid Y_{1:t+1} = y_{1:t+1}) &= \frac{p(Y_{t+1} = y_{y+1} \mid X_{t+1} = 1) p(X_{t+1} = 1 \mid Y_{1:t} = y_{1:t})}{\sum_{X_{t+1}} p(Y_{t+1} = y_{y+1} \mid X_{t+1} = 1) p(X_{t+1} = 1 \mid Y_{1:t} = y_{1:t})} \\ &= \frac{p(Y_{t+1} = y_{y+1} \mid X_{t+1} = 1) T(2,:) \pi_t}{\sum_{X_{t+1}} p(Y_{t+1} = y_{y+1} \mid X_{t+1} = 1) p(X_{t+1} = 1 \mid Y_{1:t} = y_{1:t})} \end{aligned}$$

   where $T(1,:)$, $T(2,:)$ are the first and second row in matrix T. Combine these results in matrix form, we have:

$$\pi_{t+1} \propto M(y_{t+1}) T \pi_t$$

   Normalization term: $e^T M(y_{t+1}) T \pi_t$, where e is column vector with all elements equal to one. As a result,

$$\pi_{t+1} = \frac{M(y_{t+1}) T \pi_t}{e^T M(y_{t+1}) T \pi_t}.$$

   (b) [10 Points.] On the 99th day, if the person has had 98 negative tests, with a positive test on day 99, what is the probability they have the illness? Assume the person started healthy with certainty $\pi_0 = [1 \quad 0]^T$. (Hint: write a short script to compute this.)

   Solution:

   Using data from previous questions, transition matrix and measurement likelihood matrix are:

$$T = \begin{bmatrix} 0.999 & 0.05 \\ 1 - 0.999 & 1 - 0.05 \end{bmatrix}$$

$$M(y_t = 0) = \begin{bmatrix} 0.9 & 0 \\ 0 & 0.01 \end{bmatrix}$$

$$M(y_t = 1) = \begin{bmatrix} 0.1 & 0 \\ 0 & 0.99 \end{bmatrix}$$

The probability that the student has the illness after the 99th test is $p(X_{99} = 1 \mid Y_{1:99} = y_{1:99}) = 9.916 \times 10^{-3}$.